

Pre-University Math Background Needed for the Exercises

Beyond the module's own theory (induction, complex numbers), the exercise solutions lean on math you're expected to already know from high school. Organized by knowledge area — each point now includes a short **theory** explanation, **why it's needed**, and a brief **example**.

1. Basic Algebra

1.1 Distributive property / expanding polynomials

Theory: The distributive property states $a(b+c) = ab+ac$. When multiplying two binomials $(a+b)(c+d)$, you apply it twice: multiply every term of the first factor by every term of the second, then collect like terms. **Why needed:** multiplying complex numbers (and quadratics like $p(x) \cdot q(x)$) requires "multiply everything by everything." **Example:** $(a+bx)(c+dx) = ac + adx + bcx + bdx^2$

1.2 Factoring

Theory: Factoring reverses expansion: it rewrites a sum as a product, usually by pulling out a common term shared by every summand. Recognizing a common factor lets you simplify or regroup an expression. **Why needed:** several induction proofs (e.g., $n(n+1)+2(n+1)$) require pulling out a common factor to reach the target expression. **Example:** $n(n+1) + 2(n+1) = (n+1)(n+2)$

1.3 Operating with fractions (common denominator)

Theory: Two fractions can only be added or subtracted directly once they share the same denominator. You find a common denominator (often the product of the two, or their LCM), rewrite each fraction equivalently, then add the numerators. **Why needed:** combining terms like $n/(n+1) + 1/[(n+1)(n+2)]$ into a single fraction (used constantly in induction proofs and in dividing complex numbers). **Example:** $1/2 + 1/3 = 3/6 + 2/6 = 5/6$

1.4 Solving linear equations

Theory: A linear equation ($ax+b=0$) is solved by isolating the unknown: move constants to one side and divide by the coefficient of x . It has exactly one solution whenever $a \neq 0$. **Why needed:** basic manipulation skill underlying the whole module (e.g., $x + a = 0 \rightarrow x = -a$). **Example:** $2x = 3 \rightarrow x = 3/2$

2. Quadratic Equations

2.1 The quadratic formula

Theory: Any equation of the form $ax^2+bx+c=0$ ($a \neq 0$) can be solved by "completing the square," which leads to a general formula for x in terms of a , b , and c . The formula always gives two solutions (they may coincide). **Why needed:** exercises 6 and 7 ask you to solve $ax^2+bx+c=0$ with a negative discriminant; the formula itself is assumed knowledge (the

module only tells you what to do once you get a negative number under the root). **Formula:** $x = (-b \pm \sqrt{b^2 - 4ac}) / (2a)$ **Example:** $x^2 - 2x + 5 = 0 \rightarrow x = (2 \pm \sqrt{4 - 20}) / 2 = 1 \pm 2i$

2.2 The discriminant

Theory: The discriminant $\Delta = b^2 - 4ac$ (the part under the square root in the quadratic formula) tells you the nature of the solutions without solving fully: $\Delta > 0$ gives two distinct real solutions, $\Delta = 0$ gives one repeated real solution, and $\Delta < 0$ gives no real solutions. **Why needed:** recognizing that $b^2 - 4ac < 0$ signals "no real solutions" ($\rightarrow$ complex solutions needed). **Example:** $x^2 - 6x + 25 = 0 \rightarrow \text{discriminant} = 36 - 100 = -64 < 0$

3. Exponents and Radicals

3.1 Laws of exponents

Theory: Exponents follow fixed algebraic rules regardless of what the base is: $a^m \cdot a^n = a^{m+n}$ (multiplying same base adds exponents), $(a^m)^n = a^{mn}$ (power of a power multiplies exponents), and $a^m / a^n = a^{m-n}$. These rules extend to real and complex exponents alike. **Why needed:** exponential-form operations (r^n , $e^a \cdot e^b = e^{a+b}$, $(e^a)^n = e^{an}$) rely on these rules holding for complex exponents too. **Example:** $e^{2+\pi i} = e^2 \cdot e^{\pi i}$

3.2 nth roots and simplifying radicals

Theory: The n-th root of a number x is a value y such that $y^n = x$. Radicals can be simplified by factoring out perfect powers from under the root: $\sqrt[n]{a \cdot b} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ when $a, b \geq 0$. **Why needed:** computing moduli ($\sqrt{a^2 + b^2}$) and n-th roots of complex numbers ($\sqrt[n]{r}$) requires comfort simplifying square/cube roots. **Example:** $\sqrt{20} = \sqrt{4 \cdot 5} = 2\sqrt{5}$

3.3 Rationalizing (multiply by a clever form of 1)

Theory: Multiplying a fraction by (x/x) doesn't change its value but can eliminate an inconvenient radical or imaginary unit from the denominator, since multiplying a binomial by its conjugate uses the identity $(a+b)(a-b) = a^2 - b^2$, which removes the "cross terms." **Why needed:** this is exactly the technique behind dividing complex numbers (multiply top and bottom by the conjugate) — it's the same idea as rationalizing a denominator with a radical. **Example:** $1/\sqrt{2} = \sqrt{2}/2$

4. Trigonometry

4.1 Right-triangle trigonometry (SOH-CAH-TOA)

Theory: In a right triangle, the three basic ratios relate an acute angle θ to the triangle's sides: $\sin(\theta) = \text{opposite}/\text{hypotenuse}$, $\cos(\theta) = \text{adjacent}/\text{hypotenuse}$, $\tan(\theta) = \text{opposite}/\text{adjacent} = \sin(\theta)/\cos(\theta)$. **Why needed:** converting between standard and polar form is literally "find the sides/angle of a right triangle" using sin, cos, tan. **Example:** In a right triangle, $\tan(\theta) = \text{opposite}/\text{adjacent}$

4.2 The Pythagorean theorem

Theory: In any right triangle, the square of the hypotenuse equals the sum of the squares of the two legs: $h^2 = a^2 + b^2$. It's the foundation for computing straight-line distances from perpendicular components. **Why needed:** computing the modulus $r = \sqrt{a^2 + b^2}$ is a direct application (finding a hypotenuse), and it's also used in exercise 5 to place $\sqrt{2}$ on the number line. **Example:** legs 1 and 1 $\rightarrow$ hypotenuse $= \sqrt{1^2 + 1^2} = \sqrt{2}$

4.3 Standard angle values (0° , 30° , 45° , 60° , 90° ...) and the unit circle

Theory: On a circle of radius 1 centered at the origin, the point at angle θ has coordinates $(\cos \theta, \sin \theta)$. For a handful of "nice" angles (multiples of 30° and 45°), these coordinates come out as simple numbers or simple radicals, which is why they're worth memorizing. **Why needed:** recognizing $\sin/\cos$ of "nice" angles (like $\pi/4$, $\pi/6$) lets you get exact answers instead of decimals when converting forms. **Example:** $\sin(\pi/4) = \cos(\pi/4) = \sqrt{2}/2$

4.4 Radians $\leftrightarrow$ degrees conversion

Theory: Angles can be measured in degrees (a full turn = 360°) or radians (a full turn = 2π rad, based on arc length on the unit circle). Since both measure the same full turn, 180° corresponds exactly to π radians, giving a direct conversion factor. **Why needed:** results are sometimes expected in degrees ($180^\circ = \pi$ rad), and switching between them is routine. **Formula:** degrees = radians $\cdot (180/\pi)$ **Example:** $\pi/4$ rad = 45°

4.5 Inverse trig functions (arctan) and quadrant adjustment

Theory: $\arctan(x)$ is the inverse of the tangent function, but $\tan$ repeats every π radians, so $\arctan$ can't distinguish between an angle and that angle plus π — by convention, calculators restrict $\arctan$'s output to $(-\pi/2, \pi/2)$. To recover the true angle of a point in another quadrant, you must add or subtract π based on the point's actual position (found by plotting it). **Why needed:** a calculator's $\arctan$ only returns values in $(-\pi/2, \pi/2)$, but the true angle of a complex number may lie in another quadrant — you must add/subtract π "by hand" based on where the point actually is. **Example:** for $z = -2 - 5i$ (third quadrant), $\arctan(5/2) \approx 1.19$ rad is in the *first* quadrant, so you subtract π to get the real angle: $1.19 - \pi \approx -1.95$ rad.

4.6 Trig identities for related angles (supplementary/opposite angles)

Theory: Because of the symmetry of the unit circle, the sine and cosine of an angle in quadrants 2, 3, or 4 can always be rewritten in terms of the sine/cosine of a related first-quadrant "reference angle" ($\pi - \theta$, $\pi + \theta$, or $-\theta$), with an appropriate sign flip. **Why needed:** expressing $\sin/\cos$ of angles in quadrants 2, 3, 4 in terms of a first-quadrant reference angle. **Example:** $\sin(\pi - \theta) = \sin(\theta)$; $\cos(\pi + \theta) = -\cos(\theta)$

5. Geometry

5.1 Vectors and the parallelogram law

Theory: A vector represents a displacement with both magnitude and direction, typically drawn as an arrow from the origin to a point (a,b) . Two vectors are added by placing them tail-to-tail and completing the parallelogram they form — the diagonal from the shared origin is the sum. **Why needed:** adding complex numbers geometrically is literally vector addition — place the vectors tip-to-tail or complete the parallelogram. **Example:** $(2,-1) + (1,3) = (3,2)$

5.2 Symmetry (reflections)

Theory: A reflection is a transformation that flips a point across a line (or through a point), producing its mirror image while keeping distances the same. Reflecting (a,b) across the x -axis gives $(a,-b)$; reflecting through the origin gives $(-a,-b)$. **Why needed:** the conjugate is a reflection across the x -axis; the opposite $(-z)$ is a reflection through the origin — both are basic transformation-geometry ideas. **Example:** reflecting $(2,3)$ across the x -axis gives $(2,-3)$

5.3 Compass-and-straightedge construction

Theory: Classical geometry allows constructing figures using only an unmarked ruler (straightedge) and a compass. A compass can transfer a fixed length (like a diagonal) onto another line by drawing an arc of that radius from a chosen center. **Why needed:** exercise 5 (placing $\sqrt{2}$ on the number line) is a classic classical-geometry construction: build a unit square, draw its diagonal, then swing that length onto the axis with a compass.

5.4 Regular polygons

Theory: A regular polygon has all sides and all angles equal, and its vertices lie evenly spaced on a circle — consecutive vertices are separated by the same central angle ($360^\circ/n$ for n sides). **Why needed:** recognizing that the n -th roots of a number form the vertices of a regular n -gon on a circle (exercises 28–29) draws on basic polygon/symmetry knowledge.

6. Sequences, Series, and Summation

6.1 Sigma (Σ) notation

Theory: Sigma notation is shorthand for writing a sum of many terms that follow a pattern: $\Sigma(k=1 \text{ to } n) f(k)$ means "add up $f(k)$ for every whole number k from 1 to n ." **Why needed:** sums like $1+2+\dots+n$ are written compactly as $\Sigma(k=1 \text{ to } n) k$; you need to be comfortable reading/writing this notation to follow (and write) induction proofs. **Example:** $\Sigma(k=1 \text{ to } 4) k = 1+2+3+4 = 10$

6.2 Recognizing/conjecturing a pattern from numerical data

Theory: Before a formula can be proven, it's often first *guessed* by computing several specific cases, looking for a consistent numerical pattern, and expressing that pattern algebraically as a conjecture — a hypothesis to be tested and then proven (e.g., by

induction). **Why needed:** exercise 3 gives you numeric results ($1/2, 2/3, 3/4, \dots$) and asks you to guess the general formula ($n/(n+1)$) before proving it — a basic "spot the pattern" skill from pre-algebra.

6.3 Known closed-form sums (as prior knowledge)

Theory: Certain sums of the first n terms of a simple sequence have well-known closed formulas (a single algebraic expression instead of a "... sum) — these are typically introduced in secondary-school algebra and are assumed as background facts here rather than re-derived. **Why needed:** the sum of the first n cubes, $1^3 + 2^3 + \dots + n^3 = [n(n+1)/2]^2$, is used as the very statement to be proven in exercise 1 — you're expected to recognize/accept this classic formula rather than derive it from scratch.

7. Sets and Logical Notation

7.1 Set-membership and inclusion symbols

Theory: A set is a well-defined collection of objects. " $x \in A$ " means x is a member of set A . " $A \subset B$ " means every element of A is also an element of B (A is a subset of B). These symbols give a compact, unambiguous way to state relationships between number systems. **Why needed:** statements like $N \subset Z \subset Q \subset R \subset C$ and $x \in N$ appear throughout and are foundational to how the properties are phrased. **Example:** " $3 \in N$ " reads "3 belongs to N "

7.2 Universal quantification ("for all")

Theory: The phrase "for all n " (symbol: $\forall n$) asserts that a statement holds without exception, for every single value of n in the specified set — not just for the cases you happened to check. This is a core idea in formal mathematical statements. **Why needed:** understanding that induction proves a statement "for all $n \in N$ " (not just for a few cases) is the whole logical point of the technique.

Quick-reference table

Area	Key tool	Used for
Algebra	Factoring, common denominators	Induction proofs
Quadratics	$x=(-b\pm\sqrt{(b^2-4ac)})/2a$	Finding complex roots of $ax^2+bx+c=0$
Exponents/Radicals	Exponent laws, simplifying $\sqrt{}$	Modulus, powers, exponential form
Trigonometry	SOH-CAH-TOA, Pythagoras, arctan+quadrant fix	Standard $\leftrightarrow$ polar conversion
Geometry	Vectors/parallelogram, reflections	Adding/conjugating complex numbers
Series	Σ notation, pattern-spotting	Stating & proving induction claims
Sets/Logic	$\subset, \in$, "for all n"	Framing what induction proves